# Ratio of the Sides Between a Regular Polygon and an Interior Polygon Formed by its Non-Opposite Diagonal Lines

**Deangelo Brasen,** *Stanford University*

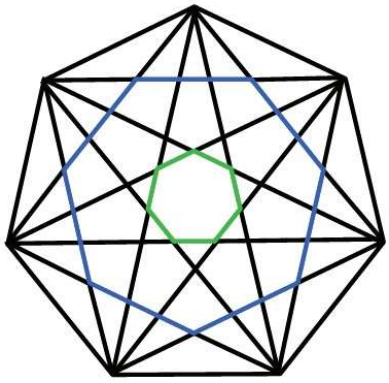
In this article, we discuss the relationship between regular polygons that are formed from intersecting, non-opposite diagonals, or diagonals that do not intercept the polygon at opposite vertices. Let the interior polygon be the polygon formed by congruent diagonal lines of a given polygon. Let the exterior polygon be the original polygon that forms the interior polygon. This proof will use properties of regular polygons and utilize the geometry of circles in relation to polygons. There will be two separate cases to analyze in this proof: the regular polygons with an odd number of sides, and regular polygons with an even number of sides. In the end, a formula will be constructed to connect the ratio of the overall original regular polygon's side length to a similar interior regular polygon's side length, based on the original polygon's diagonals. It will be shown that there will always be a defined ratio between the side length of the exterior polygon to an interior polygon.

Although not a critical formula to the overall field of mathematics, such a formula leads to a unique property of polygons never set forth before. In my time participating in mathematics competitions, I have encountered problems which can be computed very quickly with such a formula. Potential applications lead to folding in origami and materials in engineering, such as for shipping biomedical devices which require items to be compressed and blown up, or in space with the compression of materials to ensure that objects can be sent to space with minimal space taken up.

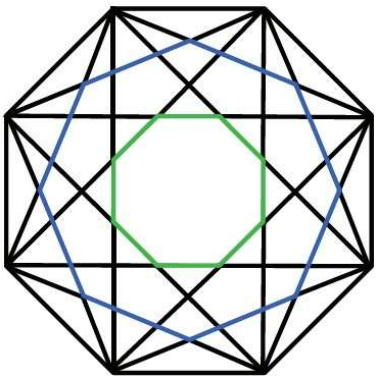
## The Problem

In order to find the ratio of side lengths between two similar regular polygons, one that is formed from drawing non-opposite diagonals between the vertices of the other polygon, we start off by looking at a part of the overall polygon.

The regular heptagon and regular octagon are shown with their diagonals drawn in Figures 1 and 2. (To view other polygons, see Appendices 1 and 2.) Each color (blue and green) represents a different smaller interior regular polygon to the original regular polygon. We can isolate one of the diagonals that form one side of the similar, interior polygon. Using this diagonal, we can use various relationships to find the desired formula—the ratio of their side lengths.



Heptagon



Octagon

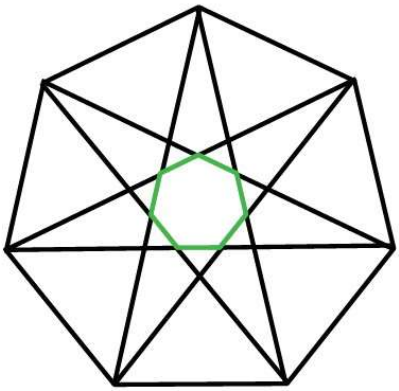
*Figure 1.* Diagrams of a heptagon and an octagon, with their non-opposite diagonals drawn to show the similar, smaller polygons, shown in blue (second smallest interior polygon) and green (smallest interior polygon).

## The Proof

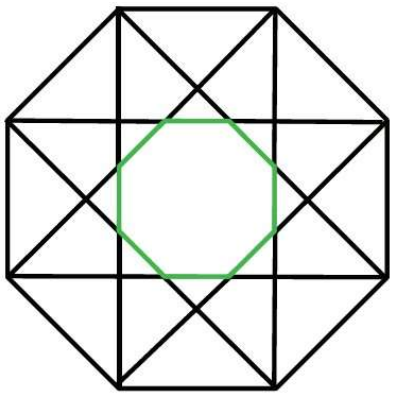
### Existence of Similar Interior Polygons

Let $\alpha$ be equal to any of the interior angles of the $n$-sided polygon: $\alpha = \pi - \frac{2\pi}{n}$. radians. Draw two arbitrary diagonal lines, each from one vertex to another, that are positioned with points being exactly one vertex away from the other diagonal. Figure 3 shows that an obtuse angle facing what would be an interior polygon forms between the diagonals at the point of intersection. Since all regular polygons are cyclic, we can

calculate the angle by inscribing this polygon in a circle. The lines form an acute angle, which appears to intercept one arc on each side of the angle. By the property of inscribed angles, the acute angle is precisely defined to be $2\pi/n$, as each arc intercepted is exactly $1/n$ of the circle. By the property of supplementary angles, the obtuse angle must be $\pi - \frac{2\pi}{n}$ or $\frac{\pi(n-2)}{n}$, which is precisely $\alpha$, the interior angle of the regular polygon.



Heptagon

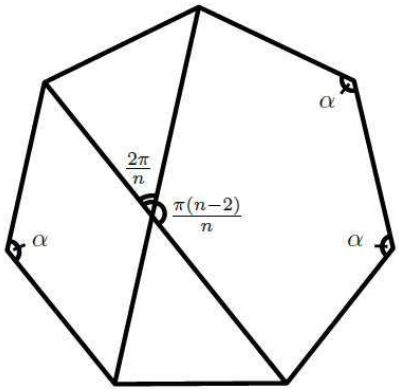


Octagon

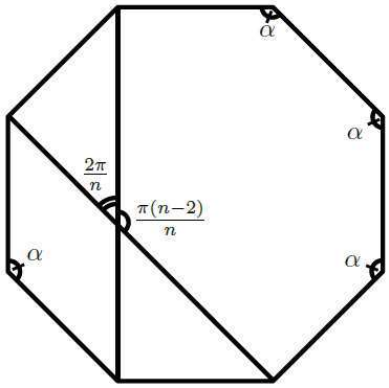
*Figure 2*. Diagrams of the smallest possible similar polygon formed from the non-opposite diagonals of a heptagon and an octagon, shown in green.

Repeating this $n$ times for each possible intersection of two arcs from two chords in a circle, we get a polygon with the sum of angles to be $\pi(n-2)$, which matches the original exterior regular polygon's sum of angles, but dilated by a certain scale factor. We also know these angles form another similar polygon, since they will form a closed shape with only angles $\alpha$.

These figures must form a closed shape, since there are only two objects that can form from these intersection lines: a shape and a point. We can restrict the possibility of a point from occurring, as that would imply that all intersecting lines intersect at only one possible point: the center of the polygon (by rotational symmetry). In addition, an $n$-sided, regular polygon has $n$ lines of rotational symmetry, which implies that there must be $n$ angles, and therefore $n$ sides, to such an interior polygon closing the figure, satisfying the criteria defining the interior polygon.




Heptagon




Octagon

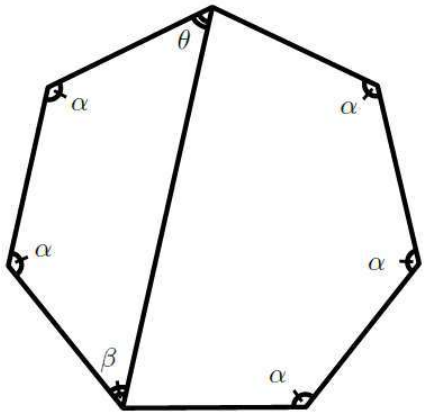
*Figure 3*. Diagrams of two diagonals of the polygon intersecting two arcs on both sides of the line form one vertex of the smaller, interior polygon.

## Proof that $\beta = \theta$

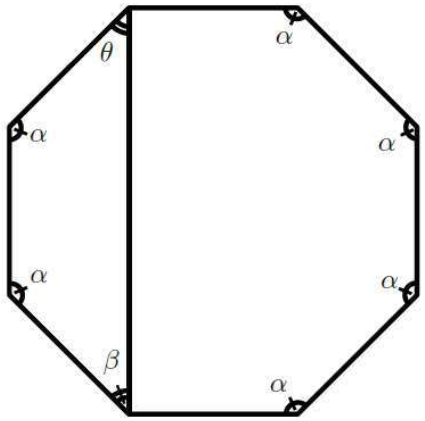
Using one of the diagonals from a side of the similar polygon, we let $\beta$ and $\theta$ be the angles formed from the difference of an overall interior angle of a vertex and the remaining angle formed from the diagonal and the smaller portion of the polygon (Figure 4). Since all regular polygons are cyclic, the polygon's vertices all lie on the circle. Since $\beta$ and $\theta$ intercept the same arc length, we must have $\beta = \theta$ by the property of inscribed angles.

## Finding $\theta$ for Polygons

To find the overall values of $\theta$, we can break the possible values into two cases: odd and even $n$-gons.




Heptagon




Octagon

*Figure 4*. Diagrams of $\beta$ and $\theta$ within the heptagon and octagon.

***Case I: Odd n***

In Figure 5, notice that the polygon's diagonals intercept one side of the polygon from a single vertex (call the angle formed $\gamma$ and the vertex the diagonals intersect at $P$). In other situations, we can visualize that the congruent diagonal lines must intersect vertices that each move one away from each other, coming from $P$. For example, the next smallest possible set of congruent diagonals would intercept three sides of the polygon, with $\gamma = 3\pi/n$. This can be generalized, leading to the largest possible number of arcs to intercept to be $n-2$ in the case where the diagonal lines are the sides itself (intercepting the nearest possible vertices from $P$). With this, we have bounds for the number of sides the diagonal lines can intersect, namely:

$$n-2 \geq 2k-1 \geq 1, \tag{1}$$

or, when solved for $k$:

$$\frac{n-1}{2} \geq k \geq 1. \tag{2}$$

Here, $k$ is some integer, and $2k-1$ is written to ensure that the value is an odd number, as the number of sides intercepted by the diagonals starts at one and increases by two for every new set of congruent diagonal lines.

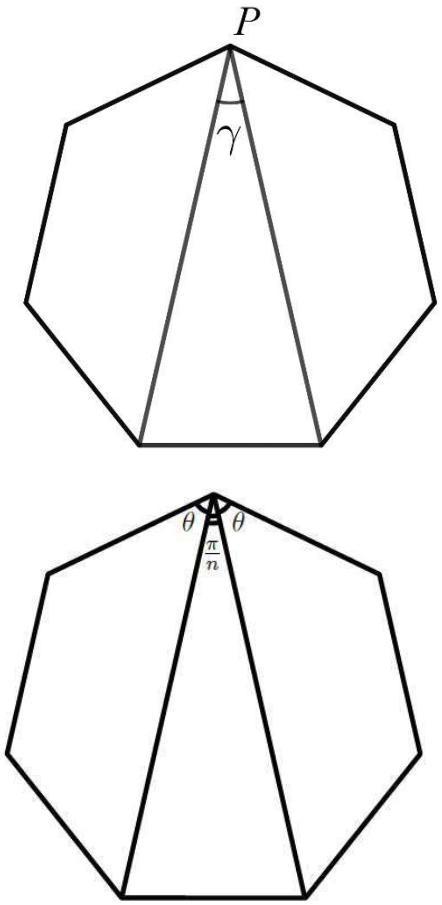



*Figure 5*. Diagram of an interior angle broken into three separate angles for an odd-sided polygon. In this case, $k = 1$ is shown in a heptagon since the diagonals intersect the polygon at one vertex ($P$) and intercept one side.

From this, we can solve for $\theta$. Since $\gamma$ intercepts $2k-1$ sides, it is exactly $\pi(2k-1)/n$. Since each side is $1/n$ of the overall circle, the central angle is exactly $2\pi/n$, and an inscribed angle of that arc would be exactly half of the central angle, $\pi/n$. We can now make an equation to find $\theta$:

$$\alpha = 2\theta + \frac{\pi(2k-1)}{n}. \tag{3}$$

Substituting $\alpha$ in terms of $n$ gives

$$\pi - \frac{2\pi}{n} = 2\theta + \frac{\pi(2k-1)}{n}. \tag{4}$$

Solving for $\theta$, the equation becomes

$$\theta = \frac{\pi}{2} - \frac{\pi(2k+1)}{2n}. \tag{5}$$

***Case II: Even n***

The even-sided polygons lead to a very similar situation, except it is important to note the change in symmetry for the

congruent diagonal lines. If the diagonal lines intersected with a vertex exactly across from $P$, they are parallel and are the same line, leading to an angle of 0. This is not very useful, since, as stated previously, we do not want to use the case where all diagonal lines intersect at the center of the polygon (essentially an extraneous case). Therefore, we start off with the diagonal lines intercepting vertices that are exactly one away from the opposite vertex, as seen in Figure 6.

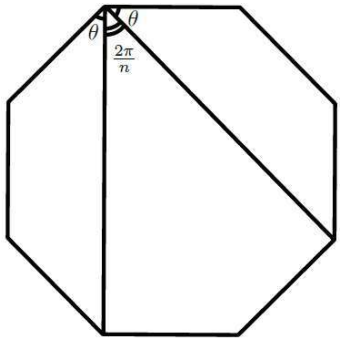



*Figure 6*. Diagram of an interior angle broken into three separate angles for an even-sided polygon In this case, $k = 1$ is shown in an octagon, since the diagonals intersect the polygon at one vertex and two other vertices that are spaced two sides apart.

Progressing as with the odd-sided polygons, we notice that the next case would be for the diagonal lines to intersect four sides of the polygon, with an angle of $4\pi/n$. This continues, with the maximum number of sides the diagonals can intersect becoming $n-2$, where the diagonals are the sides of the polygon extending from $P$. This gives us a bound for the number of sides the diagonal lines can intersect, namely

$$n-2 \geq 2k \geq 2, \tag{6}$$

or, when solved for $k$:

$$\frac{n}{2}-1 \geq k \geq 1. \tag{7}$$

Here, $k$ is some integer, and $2k$ is written to ensure that the value is even, as the number of sides intercepted by the diagonals starts at two and increases by two for every new set of congruent diagonal lines.

From this, we can again solve for $\theta$, as with the odd-sided polygons. Since $\gamma$ intercepts $2k$ sides, it is exactly $2\pi k/n$, as with the odd-sided polygons. This allows us to form the equation

$$\alpha = 2\theta + \frac{2\pi k}{n}. \tag{8}$$

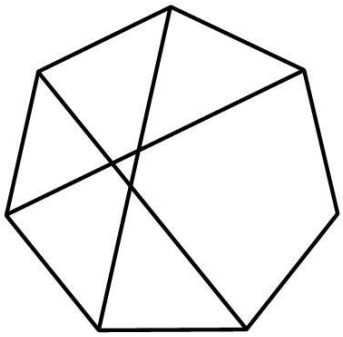
Since $\alpha = \pi - \frac{2\pi}{n}$, we now have

$$\pi - \frac{2\pi}{n} = 2\theta + \frac{2\pi k}{n}. \tag{9}$$

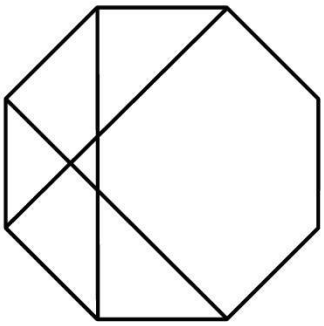
Solving for $\theta$ in terms of $n$ gives

$$\theta = \frac{\pi}{2} - \frac{\pi(1+k)}{n}. \tag{10}$$

For both odd and even $n$, we can define $k$ as the "level of compression," which represents each possible interior polygon that can form from the diagonals. With $k = 1$, the smallest possible polygon forms: We obtain the green polygons in Figure 1. Likewise, $k = 2$ represents the blue polygons, and so forth. In the ultimate case (as illustrated in the heptagon and octagon of Figure 2, the largest possible interior polygon is technically the polygon itself, or $k = 3$ for both. An interesting connection is that consecutive odd- and even-sided polygons have the same number of levels of compression. For a given integer $p$ (an arbitrary integer with no significance on its own), the polygons with $2p-1$ and $2p$ sides will always have the same levels of compression (in the case of the examples in the figures, $p = 4$ was used).



Heptagon



Octagon

*Figure 7*. Three congruent lines within the heptagon and octagon are shown.

Appendix 2 displays the first compression level for regular polygons with 3 to 8 sides of length $x$ for the interior polygon. Note the changing bounds of $k$ as the number of sides of a polygon increases by two. In particular, $p$ is always one more than the maximum value of $k$. In the case $p = 2$, (see Appendix 1) the largest possible $k$ is one. When $p = 3$, the largest possible $k$ is two. This pattern continues with $p = 4$, as mentioned previously. For every increase in $p$ by one, there is another "level of compression" that can occur, because the

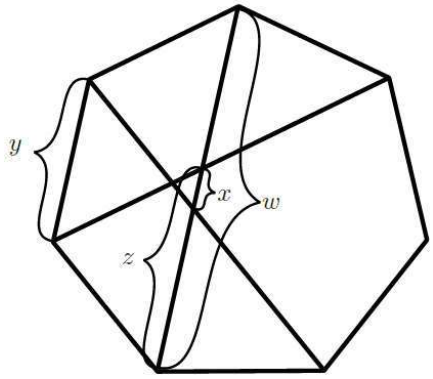
upper bound for the number of sides intersected by two congruent diagonals increases by two. Therefore, the maximum value of $k$ increases by one (based on Equations (2) and (7)).

We can use this fact to combine the inequalities of both the odd- and even-sided polygons: The maximum $k$ is $p-1$. Therefore,

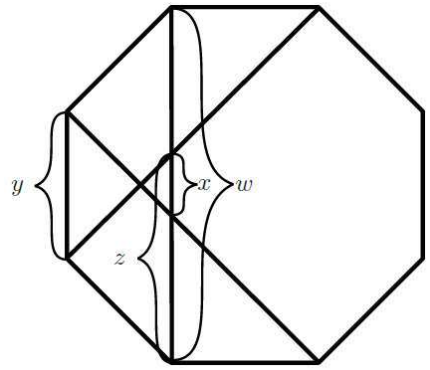
$$p-1 \geq k \geq 1. \tag{11}$$

## The General Formula

Now, we generalize the ratio of side lengths for any level of compression, using the case where $k = 1$, and $p = 4$ to help




Heptagon



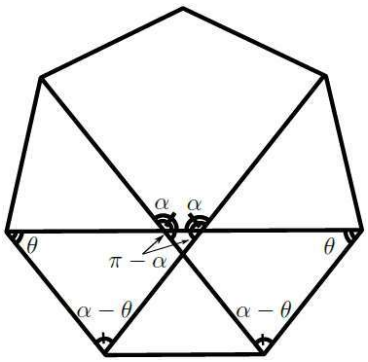

Octagon

*Figure 8*. Diagrams of the heptagon and octagon with lengths $w$, $x$, $y$, and $z$ shown.

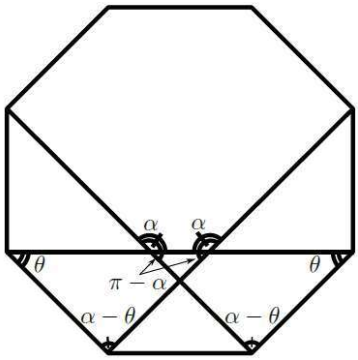
visualize the situation. Figure 7 shows the lines used to form a side of the interior polygon with two vertices. Figure 8 labels various lengths that will determine the ratio, namely $w$, $x$, $y$, and $z$. Our goal is to find $x/y$.

We can now use the Law of Sines to solve for the smaller side. Let $x$ be the length of the interior polygon for a specific diagonal, $y$ be the length of the exterior polygon, $z$ be the length of the diagonal up to the farther side of the interior polygon, and $w$ be the length of the entire diagonal, as shown in Figure 8.

We can use two triangles to solve for the overall formula. First, we use a triangle with sides $z$ and $y$, and the other with sides $w$ and $y$.




Heptagon




Octagon

*Figure 9*. Diagrams of the heptagon and octagon with specified angles.

### The Case of the Triangle with Sides *z* and *y*

We know that one of the angles is $\theta$, but we also need to know the other two angles to use the Law of Sines. By rotational symmetry, we can draw two diagonals that are congruent to the original one but rotated one vertex to the right or left (Figure 9). We have already shown that the obtuse angle formed by the intersection of the diagonals must be $\alpha$, since it forms the sides of the interior polygon. Therefore, the acute angle formed must be $\pi - \alpha$. Using the property that angles in any triangle sum up to $\pi$, we see that the final angle is $\alpha - \theta$, as labeled in Figure 9.

Using the Law of Sines, we have

$$\frac{\sin(\alpha-\theta)}{z} = \frac{\sin(\pi-\alpha)}{y}. \tag{12}$$

Solving for $z$ in terms of $y$ gives

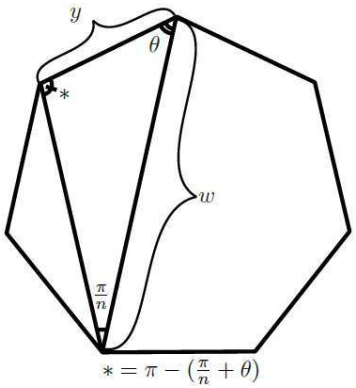
$$z = \frac{y\sin(\alpha-\theta)}{\sin(\pi-\alpha)}. \tag{13}$$

Since $\sin(\pi-\alpha) = \sin(\alpha)$, the equation simplifies to:

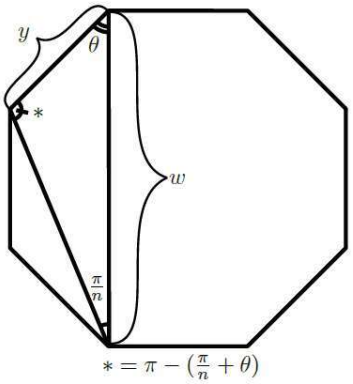
$$z = \frac{y \sin(\alpha - \theta)}{\sin \alpha}. \tag{14}$$

## The Case of the Triangle with Sides *w* and *y*

We can now use the second triangle with sides $w$ and $y$ to find $w$ (see Figure 10). Since one of the angles is $\theta$ and the other is an inscribed angle intercepting one arc, we see that the angles of the triangle are $\theta$, $\pi/n$, and $\pi - \left(\frac{\pi}{n} + \theta\right)$.




Heptagon




Octagon

*Figure 10*. Diagrams of the triangle with lengths $w$ and $y$ within the heptagon and octagon with their angles shown.

With this, we can use the Law of Sines again:

$$\frac{\sin\left(\frac{\pi}{n}\right)}{y} = \frac{\sin\left(\pi - \left(\frac{\pi}{n} + \theta\right)\right)}{w}. \tag{15}$$

Solving for $w$ in terms of $y$ gives us the equation:

$$w = \frac{y \sin\left(\pi - \left(\frac{\pi}{n} + \theta\right)\right)}{\sin\left(\frac{\pi}{n}\right)}. \tag{16}$$

## The Overall Formula

The key idea of the proof is to notice an overlap in the triangles of sides $z$ and $y$. When both sides of length $z$ are added together, the sum geometrically represents $w$ with an extra length $x$, since that is where the sides of the triangles overlap along $w$ (see Figure 8), with the sides labeled by their respective variables. This provides the following formula to relate $z$, $x$, and $w$:

$$2z - x = w. \tag{17}$$

Substituting in $z$ and $w$ gives the equation

$$2\left(\frac{y \sin(\alpha - \theta)}{\sin \alpha}\right) - x = \frac{y \sin\left(\pi - \left(\frac{\pi}{n} + \theta\right)\right)}{\sin\left(\frac{\pi}{n}\right)}. \tag{18}$$

Solving for $x$ in terms of $y$ gives the equation

$$x = y\left[\frac{2\sin(\alpha - \theta)}{\sin \alpha} - \frac{\sin\left(\pi - \left(\frac{\pi}{n} + \theta\right)\right)}{\sin\left(\frac{\pi}{n}\right)}\right]. \tag{19}$$

Therefore, the ratio of the side length of the interior polygon to the side length of the exterior polygon is:

$$\frac{x}{y} = \frac{2\sin(\alpha - \theta)}{\sin \alpha} - \frac{\sin\left(\pi - \left(\frac{\pi}{n} + \theta\right)\right)}{\sin\left(\frac{\pi}{n}\right)}. \tag{20}$$

Since $\alpha = \pi - 2\pi/n$, we have

$$\frac{x}{y} = \frac{2\sin\left(\pi - \left(\frac{2\pi}{n} + \theta\right)\right)}{\sin\left(\pi - \frac{2\pi}{n}\right)} - \frac{\sin\left(\pi - \left(\frac{\pi}{n} + \theta\right)\right)}{\sin\left(\frac{\pi}{n}\right)}. \tag{21}$$

Since $\sin(\pi - m) = \sin(m)$, the equation above becomes

$$\frac{x}{y} = \frac{2\sin\left(\frac{2\pi}{n} + \theta\right)}{\sin\left(\frac{2\pi}{n}\right)} - \frac{\sin\left(\frac{\pi}{n} + \theta\right)}{\sin\left(\frac{\pi}{n}\right)}$$

or, equivalently,

$$\frac{x}{y} = \frac{2\sin\left(\left[\frac{\pi}{n} + \theta\right] + \frac{\pi}{n}\right)}{\sin\left(\frac{2\pi}{n}\right)} - \frac{\sin\left(\frac{\pi}{n} + \theta\right)}{\sin\left(\frac{\pi}{n}\right)}. \tag{22}$$

This can be simplified further by using the double angle identity $\sin(2\beta) = 2\sin(\beta)\cos(\beta)$, along with the sum identity $\sin(\beta + \gamma) = \sin(\beta)\cos(\gamma) + \cos(\beta)\sin(\gamma)$ on the denominator and numerator of the first fraction, respectively.

The result is $\frac{x}{y} =$

$$\frac{2\left[\sin\left(\frac{\pi}{n}+\theta\right)\cos\left(\frac{\pi}{n}\right)+\sin\left(\frac{\pi}{n}\right)\cos\left(\frac{\pi}{n}+\theta\right)\right]}{2\sin\left(\frac{\pi}{n}\right)\cos\left(\frac{\pi}{n}\right)}-\frac{\sin\left(\frac{\pi}{n}+\theta\right)}{\sin\left(\frac{\pi}{n}\right)}.$$

Splitting and simplifying the first fraction provides

$$\frac{x}{y}=\frac{\sin\left(\frac{\pi}{n}+\theta\right)}{\sin\left(\frac{\pi}{n}\right)}+\frac{\cos\left(\frac{\pi}{n}+\theta\right)}{\cos\left(\frac{\pi}{n}\right)}-\frac{\sin\left(\frac{\pi}{n}+\theta\right)}{\sin\left(\frac{\pi}{n}\right)} \tag{23}$$

$$\Rightarrow \quad \frac{x}{y}=\frac{\cos\left(\frac{\pi}{n}+\theta\right)}{\cos\left(\frac{\pi}{n}\right)}. \tag{24}$$

All angles are measured in radians, and $\theta$ varies whether the polygon is even or odd: $\theta=\frac{\pi}{2}-\frac{\pi(1+k)}{n}$ for even-sided polygons and $\theta=\frac{\pi}{2}-\frac{\pi(2k+1)}{2n}$ for odd-sided polygons, for integers $k$, such that $p-1\geq k\geq 1$.

### Alternative Formulas and Simplifying $\theta$

Although not a crucial aspect of the proof, this section does provide unique ways to memorize and compute the formula, which may be useful to mathematics competitors or just as a way to easily memorize and compute the formula.

An alternative equation can be seen using the cosine sum identity $\cos(\beta+\gamma)=\cos(\beta)\cos(\gamma)-\sin(\beta)\sin(\gamma)$ to provide the following simplification:

$$\frac{x}{y}=\frac{\cos\left(\frac{\pi}{n}\right)\cos(\theta)-\sin\left(\frac{\pi}{n}\right)\sin(\theta)}{\cos\left(\frac{\pi}{n}\right)} \tag{25}$$

$$\Rightarrow \quad \frac{x}{y}=\cos(\theta)-\tan\left(\frac{\pi}{n}\right)\sin(\theta), \tag{26}$$

with $\theta=\frac{\pi}{2}-\frac{\pi(1+k)}{n}$ for even-sided polygons and $\theta=\frac{\pi}{2}-\frac{\pi(2k+1)}{2n}$ for odd-sided polygons, and $p-1\geq k\geq 1$.

By applying the substitution $\theta'=\pi/2-\theta$ and the formula $\sin(\beta-\gamma)=\sin(\beta)\cos(\gamma)-\sin(\gamma)\cos(\beta)$, the following formulas can be used instead of the cofunction identities for sine and cosine:

$$\frac{x}{y}=\sin(\theta')-\tan\left(\frac{\pi}{n}\right)\cos(\theta') \tag{27}$$

$$=\frac{\sin(\theta')\cos\left(\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right)}-\frac{\sin\left(\frac{\pi}{n}\right)\cos(\theta')}{\cos\left(\frac{\pi}{n}\right)}$$

$$=\frac{\sin(\theta')\cos\left(\frac{\pi}{n}\right)-\sin\left(\frac{\pi}{n}\right)\cos(\theta')}{\cos\left(\frac{\pi}{n}\right)}$$

$$\Rightarrow \quad \frac{x}{y}=\frac{\sin\left(\theta'-\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right)}, \tag{28}$$

with $\theta'=\frac{\pi(1+k)}{n}$ for even-sided polygons, $\theta'=\frac{\pi(2k+1)}{2n}$ for odd-sided polygons, and $p-1\geq k\geq 1$.

## Conclusion

In the end, the foregoing formulas establish that the hypothesis is, indeed, supported, in that every polygon among every possible "level of compression" has a unique ratio between the side lengths of the interior polygon and the exterior polygon. A visual for polygons with interior polygons defined by $k=1$ for the first six regular polygons are shown in Appendix 2. The data table in Appendix 1 supports the fact that the formula closely matches hand-measured values of $x/y$. Although the measured values can lead to numerous sources of error, the margin of error is remarkably low.



**Deangelo Brasen** is a freshman undergraduate at Stanford University. Currently, his plans are to major in math and physics. He has been learning calculus, analysis, and linear algebra, with interests in relativity and quantum mechanics.

# Appendices: Ratio of the Sides between a Regular Polygon and an Interior Polygon Formed by its Non-Opposite Diagonal Lines

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## Appendix 1: Data Values for the First Six Regular Polygons

Let $\phi$ be the Golden Ratio, $\left(1+\sqrt{5}\right)/2$.

**Table 1**

*Exact/Approximate Values of the Side Length Ratio Between Regular and Compressed Polygons*

| Compression Level | The Type of Regular Polygon | | | | | |
|---|---|---|---|---|---|---|
| | Triangle | Square | Pentagon | Hexagon | Heptagon | Octagon |
| $k = 1$ | 1 | 1 | $2-\phi$ $\approx 0.382$ | $\sqrt{3}/3$ $\approx 0.577$ | $4\cos^2(\pi/7)-3$ $\approx 0.247$ | $\sqrt{2}-1$ $\approx 0.414$ |
| $k = 2$ | DNE* | DNE* | 1 | 1 | $2\cos(\pi/7)-\sec(\pi/7)$ $\approx 0.692$ | |
| $k = 3$ | DNE* | DNE* | DNE* | DNE* | 1 | 1 |

*Note.* *Angle is out of boundaries.

**Table 2**

*Measured Values of the Side Length Ratio Between Regular and Compressed Polygons*

| Compression Level | The Type of Regular Polygon | | | | | |
|---|---|---|---|---|---|---|
| | Triangle | Square | Pentagon | Hexagon | Heptagon | Octagon |
| $k = 1$ | 1 | 1 | 0.4 | 0.5624 | 0.256 | 0.4 |
| $k = 2$ | DNE* | DNE* | 1 | 1 | 0.733 | 0.8 |
| $k = 3$ | DNE* | DNE* | DNE* | DNE* | 1 | 1 |

*Note.* *Angle is out of boundaries.



## Appendix 2: Diagrams of the First Six Regular Polygons with their Interior Polygon's First Compression Level

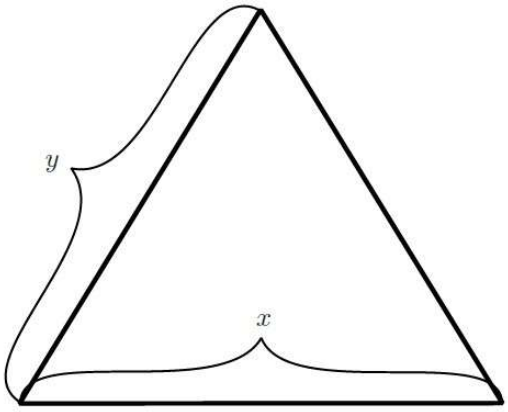



*Figure 11*. Triangle. Measured $\frac{x}{y}=1$; calculated $\frac{x}{y}=1$.

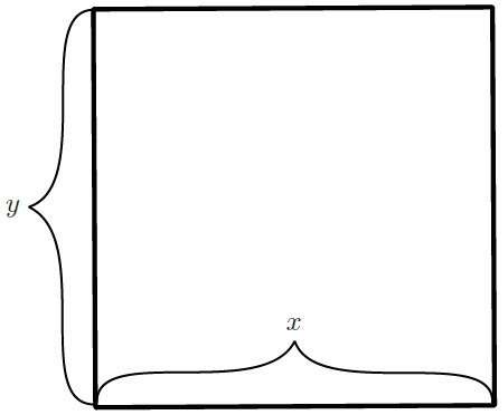



*Figure 12*. Square. Measured $\frac{x}{y}=1$; calculated $\frac{x}{y}=1$.

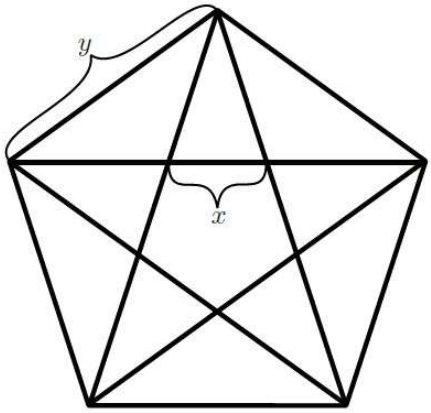



*Figure 13*. Pentagon. Measured $\frac{x}{y}=0.4$; calculated $\frac{x}{y}=2-\phi\approx 0.382$.

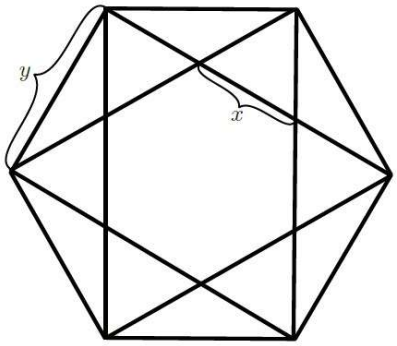



*Figure 14*. Hexagon. Measured $\frac{x}{y}=0.5624$; calculated $\frac{x}{y}=\frac{\sqrt{3}}{3}\approx 0.577$.

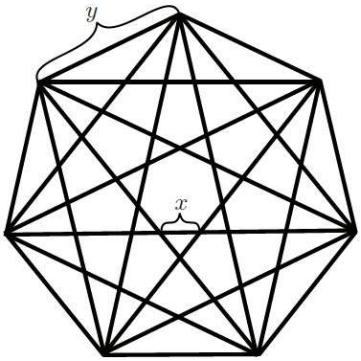



*Figure 15*. Septagon. Measured: $\frac{x}{y}=0.256$; calculated $\frac{x}{y}=4\cos^2\left(\frac{\pi}{7}\right)\approx 0.247$.

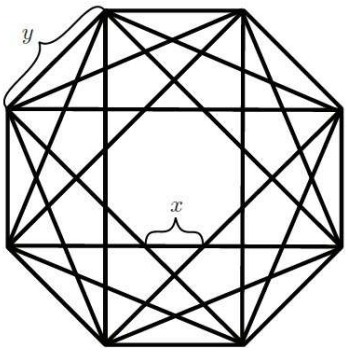



*Figure 16*. Octagon. Measured $\frac{x}{y}=0.4$; calculated $\frac{x}{y}=\sqrt{2}-1\approx 0.414$.